AI + ROBOTICS + AUTOMATION

SMART AI TRASH CAN

Auto Interceptor, Compactor & Sealing Bin

A Fully Conversational AI-Powered Waste Management Robot

- AI Segregation
- Auto Interceptor
- Voice Control
- Auto Compaction
- Auto Bag Sealing
- Fill-Level Monitoring
- Hands-Free

Project Presentation • 2026

SUSTAINABILITY ALIGNMENT

Sustainable Development Goals

SDG 3

Good Health & Well-Being

- Reduces direct contact with waste
- Improves hygiene through touch-free operation
- Reduces exposure to unpleasant waste

SDG 11

Sustainable Cities & Communities

- Cleaner, smarter waste management
- Prevents overflowing bins & litter
- Technology-driven urban sanitation

SDG 12

Responsible Consumption & Production

- Encourages proper waste segregation
- Separates biodegradable & non-biodegradable waste
- Promotes efficient waste handling

PROJECT CONTRIBUTION: AI + Automation + Hygiene + Responsible Waste Management

Existing Waste Management Problems

Improper Segregation

Biodegradable and non-biodegradable waste often gets mixed.

Unhygienic Contact

Users touch bins and lids while disposing or removing waste.

Overflow

Conventional bins give no awareness of fill level.

Manual Disposal

Users must physically approach and operate the bin.

No Compaction

Uncompacted waste occupies unnecessary space.

Bag Removal

Removing full bags requires contact with the bin.

No Interaction

Traditional bins provide no real-time info or guidance.

THE NEED: A smart, touch-free, automated waste-management system combining AI, robotics & conversational interaction.

THE INNOVATION

Smart AI Trash Can — Core Solution

AI Vision

Identifies waste; detects user gestures & objects.

Auto Interceptor

Moves toward the user or predicted trash location.

Auto Segregation

Separates biodegradable & non-biodegradable waste.

Conversational AI

Understands commands, questions & casual chat.

Auto Compaction

Compresses waste to increase usable capacity.

Auto Bag Sealing

Seals the garbage bag when the bin reaches its limit.

Smart Monitoring

Tracks fill level and provides alerts.

Touch-Free

Minimizes physical contact throughout disposal.

ONE-LINE CONCEPT

“The bin comes to you, catches your waste, sorts it, compacts it, seals it — and tells you when it’s ready to be emptied.”

WHAT WE AIM TO ACHIEVE

Project Objectives

1. Automate waste segregation using AI-based waste identification
2. Enable touch-free disposal through automatic lid control
3. Intercept thrown waste using camera-based object tracking
4. Approach users automatically on a raised-hand gesture
5. Provide conversational interaction via voice commands & AI
6. Monitor waste levels using ultrasonic & other sensors
7. Compact waste automatically to improve bin capacity
8. Seal full garbage bags automatically for cleaner handling
9. Give real-time alerts when the bag is ready for removal
10. Build a mobile, intelligent system for homes & smart environments

SYSTEM ARCHITECTURE

System Workflow

USER / WASTE

- Camera + Microphone detect waste
- Hand-gesture detection
- Thrown-object tracking
- Voice commands

AI PROCESSING

- Object detection
- Waste classification
- Gesture recognition
- Voice-intent recognition

CENTRAL CONTROLLER

- ESP32 / Raspberry Pi
- Processes sensor data
- Controls the full system

ACTION LAYER

- Motor movement
- Lid open / close
- Waste segregation
- Compaction
- Bag sealing

MONITORING

- Ultrasonic fill detection
- IR bag detection
- MQ-135 gas / smell
- LED + buzzer alerts
- Phone notification

SMART BIN RETURNS TO IDLE STATE

Interaction Modes: **Catch Mode** — tracks thrown waste & moves to intercept it **Service Mode** — detects a raised hand & approaches the user

Hardware & Software Components

ESP32
Main microcontroller & system control

ESP32-CAM
Object & gesture detection

Raspberry Pi 4
AI / conversational processing

HC-SR04 Ultrasonic
Fill-level & obstacle detection

IR Sensor
Bag / fullness confirmation

DC Gear Motors + L298N
Robotic movement & control

MG996R Servo
Automatic lid control

12V Linear Actuator + Press Plate
Waste compaction

Auto Bag Sealing Unit
Seals & cuts garbage bags

Mic + Speaker + Amp
Voice input & responses

MQ-135 Gas Sensor
Air-quality / odor detection

LED Ring + Buzzer
Status & warning alerts

12V Li-ion Battery
Portable power supply

ABS Body + Silicone Gaskets
Water-resistant structure

SOFTWARE STACK: Arduino IDE • Computer Vision • Object Detection • Speech-to-Text • LLM • Text-to-Speech

PROJECT ECONOMICS

Estimated Budget

Raspberry Pi 4	₹4,000 – ₹6,500
Cameras / Sensors	₹1,000 – ₹2,500
Ultrasonic + IR Sensors	₹400 – ₹800
DC Motors + Wheels	₹700 – ₹1,500
Bag Sealing Mechanism	₹1,000 – ₹2,500
Chassis + Smart Bin Body	₹1,500 – ₹3,000
ESP32 + ESP32-CAM	₹800 – ₹1,500
Microphone + Speaker	₹800 – ₹1,800
Servo + Motor Driver	₹700 – ₹1,600
Linear Actuator + Press Plate	₹1,500 – ₹3,000
Battery + Power System	₹1,500 – ₹3,000
Other Components & Wiring	₹1,000 – ₹2,000

ESTIMATED TOTAL: ₹14,900 – ₹29,700

Cost varies with component quality, AI hardware & prototype size.

WHAT'S NEXT

Future Enhancements

Advanced AI Classification

Identify plastic, paper, metal, glass & food waste.

Autonomous Navigation

Advanced mapping & navigation for indoor spaces.

Mobile Application

View fill levels, stats, alerts & status remotely.

IoT & Cloud Dashboard

Store waste-generation data & generate analytics.

Automatic Charging

Returns to a charging station when battery is low.

Improved Compaction

Optimize compaction by waste type & available space.

Smart Building Integration

Connect bins across homes, offices, hospitals & campuses.

Waste Analytics

Analyze disposal patterns to build better habits.

LONG-TERM VISION: A network of intelligent waste-management robots working together in smart homes & smart cities.

Conclusion

The Smart AI Trash Can unifies **AI, robotics, computer vision, sensors & conversational tech** into one waste-management system that can:

- Identify & segregate waste
- Move toward users or thrown waste
- Understand voice commands
- Compact waste automatically
- Seal full garbage bags
- Monitor waste levels
- Alert users when needed
- Reduce physical contact

FINAL MESSAGE

"From simply holding waste to intelligently managing it."

AI + Robotics + Automation → Smarter, Cleaner & More Hygienic Waste Management

THANK YOU • QUESTIONS?